

# Case 1: WebShop Footwear (Functional Constraints)

**Task:** I want a pair of men's black slip-resistant work shoes, size 10, rubber sole, price lower than \$50.00.

|               |                                                                                            |
|---------------|--------------------------------------------------------------------------------------------|
| <b>Skills</b> | [foo_002] Verify features in description; [foo_004] Confirm price after variant selection. |
| <b>Avoid</b>  | [err_001] Omitting price cap; [err_003] Selecting wrong product category.                  |

## Agent Trajectory with Skill-Guided Reasoning

|               |                                                                                                                                                                                                                                                |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Step 1</b> | <b>Obs:</b> WebShop homepage.<br><b>Think:</b> Per [gen_001], I'll include key constraints (type, function, color, price) in the query.<br><b>Action:</b> search[men black slip resistant work shoes rubber sole under 50]                     |
| <b>Step 2</b> | <b>Obs:</b> Results: [1] Chef Shoes \$38.99; [2] Kitchen Shoes \$42.50; [3] Steel Toe Boots \$55.99.<br><b>Think:</b> [gen_003] Item [1] is under budget and matches function. Item [3] is over budget (skip).<br><b>Action:</b> click[Item 1] |
| <b>Step 3</b> | <b>Obs:</b> Product page. Price: \$35.99-\$45.99. Colors: Black, White. Sizes: 7-12.<br><b>Think:</b> [foo_002] Description confirms rubber sole. Now selecting Black and Size 10 [foo_003].<br><b>Action:</b> click[Black] → click[10]        |
| <b>Step 5</b> | <b>Obs:</b> Selected: Black, Size 10. Price updated to \$38.99.<br><b>Think:</b> [foo_004] \$38.99 ≤ \$50. All constraints (color, size, function, price) are satisfied.<br><b>Action:</b> click[Buy Now]                                      |

**Result: SUCCESS** — Purchased at \$38.99. **Key Skills:** [foo\_002] (Verification), [foo\_004] (Price Check).